

Inequalities to Interval Notation: Detailed Solutions

For each problem: (i) solve the inequality, (ii) write the interval form, (iii) interpret it on a number line.

1 Core Rules

- Adding/subtracting the same number on both sides does *not* change the inequality direction.
- Multiplying/dividing by a **positive** number does *not* change the inequality direction.
- Multiplying/dividing by a **negative** number **flips** the inequality sign.
- Endpoints: $<$ or $>$ means **open** endpoint ($)$; $\leq$ or $\geq$ means **closed** endpoint [$]$.

2 Solve Each Inequality and Write the Solution in Interval Notation

Worked Example 1

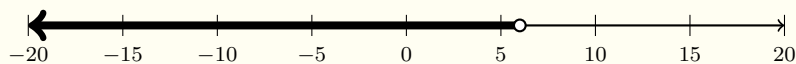
Solve $3x - 7 < 11$.

Step 1 (algebra).

$$3x - 7 < 11 \Rightarrow 3x < 18 \Rightarrow x < 6.$$

Step 2 (interval form). $(-\infty, 6)$.

Step 3 (number line).



Worked Example 2

Solve $5 - 2x \leq 19$.

Step 1 (algebra).

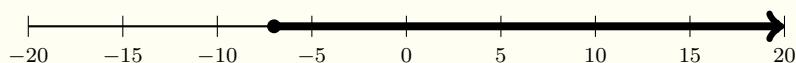
$$5 - 2x \leq 19 \Rightarrow -2x \leq 14.$$

Divide by -2 (negative), so **flip** the inequality:

$$x \geq -7.$$

Step 2 (interval form). $[-7, \infty)$.

Step 3 (number line).



Worked Example 3

Solve $-4x + 9 \geq -3$.

Step 1 (algebra).

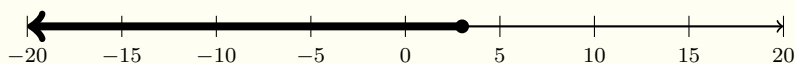
$$-4x + 9 \geq -3 \Rightarrow -4x \geq -12.$$

Divide by -4 (negative), so **flip**:

$$x \leq 3.$$

Step 2 (interval form). $(-\infty, 3]$.

Step 3 (number line).



Worked Example 4

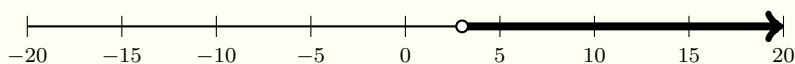
Solve $7x + 1 > 22$.

Step 1 (algebra).

$$7x + 1 > 22 \Rightarrow 7x > 21 \Rightarrow x > 3.$$

Step 2 (interval form). $(3, \infty)$.

Step 3 (number line).



Worked Example 5

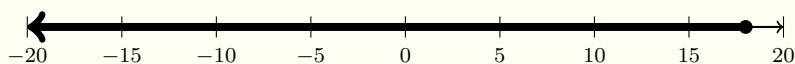
Solve $\frac{x}{3} - 4 \leq 2$.

Step 1 (algebra).

$$\frac{x}{3} - 4 \leq 2 \Rightarrow \frac{x}{3} \leq 6 \Rightarrow x \leq 18.$$

Step 2 (interval form). $(-\infty, 18]$.

Step 3 (number line).



Worked Example 6

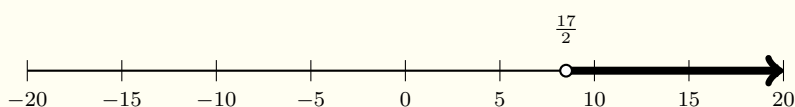
Solve $\frac{2x - 5}{4} > 3$.

Step 1 (algebra). Multiply by 4 (positive, no flip):

$$2x - 5 > 12 \Rightarrow 2x > 17 \Rightarrow x > \frac{17}{2} = 8.5.$$

Step 2 (interval form). $\left(\frac{17}{2}, \infty\right)$.

Step 3 (number line).



Worked Example 7

Solve $9 - \frac{x}{2} \geq 1$.

Step 1 (algebra).

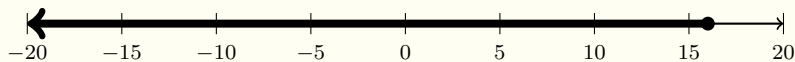
$$9 - \frac{x}{2} \geq 1 \Rightarrow -\frac{x}{2} \geq -8.$$

Multiply by -2 (negative, flip):

$$x \leq 16.$$

Step 2 (interval form). $(-\infty, 16]$.

Step 3 (number line).



Worked Example 8

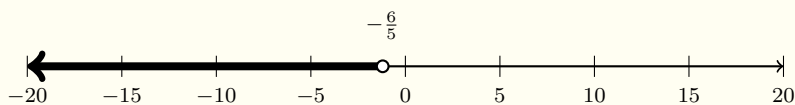
Solve $\frac{5}{2}x + 3 < 0$.

Step 1 (algebra).

$$\frac{5}{2}x + 3 < 0 \Rightarrow \frac{5}{2}x < -3 \Rightarrow x < -\frac{6}{5} = -1.2.$$

Step 2 (interval form). $\left(-\infty, -\frac{6}{5}\right)$.

Step 3 (number line).



Worked Example 9

Solve $-\frac{3}{4}x - 2 \leq 7$.

Step 1 (algebra).

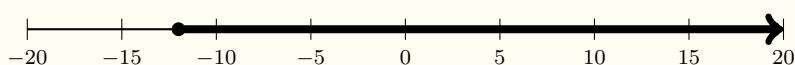
$$-\frac{3}{4}x - 2 \leq 7 \Rightarrow -\frac{3}{4}x \leq 9.$$

Divide by $-\frac{3}{4}$ (negative, flip):

$$x \geq 9 \left(-\frac{4}{3}\right) = -12.$$

Step 2 (interval form). $[-12, \infty)$.

Step 3 (number line).



Worked Example 10

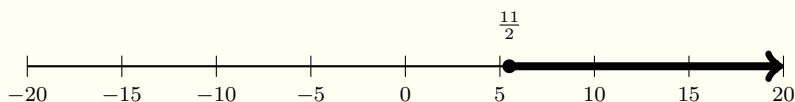
Solve $2(3x - 5) \geq 4x + 1$.

Step 1 (algebra).

$$2(3x - 5) \geq 4x + 1 \Rightarrow 6x - 10 \geq 4x + 1 \Rightarrow 2x \geq 11 \Rightarrow x \geq \frac{11}{2} = 5.5.$$

Step 2 (interval form). $\left[\frac{11}{2}, \infty\right)$.

Step 3 (number line).



Worked Example 11

Solve $4 - (x - 6) < 2x + 1$.

Step 1 (algebra). Simplify the left side: $4 - (x - 6) = 4 - x + 6 = 10 - x$. So

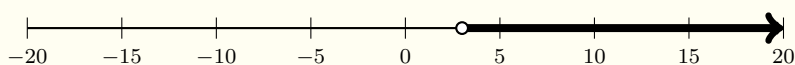
$$10 - x < 2x + 1 \Rightarrow 10 - 3x < 1 \Rightarrow -3x < -9.$$

Divide by -3 (negative, flip):

$$x > 3.$$

Step 2 (interval form). $(3, \infty)$.

Step 3 (number line).



Worked Example 12

Solve $3(x + 2) - 2(2x - 1) \leq 9$.

Step 1 (algebra). Expand:

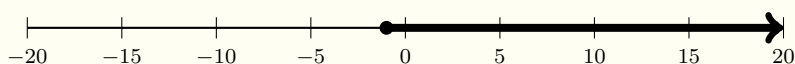
$$3(x + 2) = 3x + 6, \quad -2(2x - 1) = -(4x - 2) = -4x + 2.$$

So

$$(3x + 6) + (-4x + 2) \leq 9 \Rightarrow -x + 8 \leq 9 \Rightarrow -x \leq 1 \Rightarrow x \geq -1.$$

Step 2 (interval form). $[-1, \infty)$.

Step 3 (number line).



Worked Example 13

Solve $6 - 5(2 - x) > 3x + 4$.

Step 1 (algebra). Compute $-5(2 - x) = -10 + 5x$:

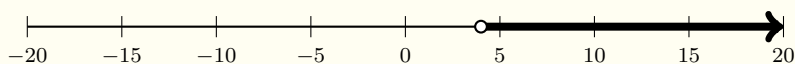
$$6 - 5(2 - x) = 6 - 10 + 5x = -4 + 5x.$$

Then

$$-4 + 5x > 3x + 4 \Rightarrow 2x > 8 \Rightarrow x > 4.$$

Step 2 (interval form). $(4, \infty)$.

Step 3 (number line).



Worked Example 14

Solve $10 - (3x + 2) \leq 2(1 - x)$.

Step 1 (algebra). Simplify:

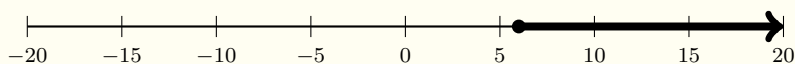
$$10 - (3x + 2) = 8 - 3x, \quad 2(1 - x) = 2 - 2x.$$

So

$$8 - 3x \leq 2 - 2x \Rightarrow 8 - x \leq 2 \Rightarrow -x \leq -6 \Rightarrow x \geq 6.$$

Step 2 (interval form). $[6, \infty)$.

Step 3 (number line).



Worked Example 15

Solve $\frac{x-1}{5} - \frac{x+4}{10} \geq 0$.

Step 1 (algebra). Use common denominator 10:

$$\frac{x-1}{5} = \frac{2(x-1)}{10}.$$

Thus

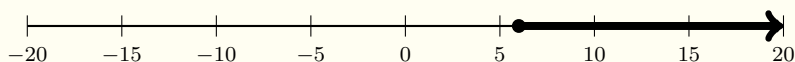
$$\frac{2(x-1) - (x+4)}{10} \geq 0 \Rightarrow \frac{x-6}{10} \geq 0.$$

Since $10 > 0$:

$$x - 6 \geq 0 \Rightarrow x \geq 6.$$

Step 2 (interval form). $[6, \infty)$.

Step 3 (number line).



Worked Example 16

Solve $\frac{3x+2}{6} + \frac{x-5}{3} < 1$.

Step 1 (algebra). Write $\frac{x-5}{3} = \frac{2(x-5)}{6}$:

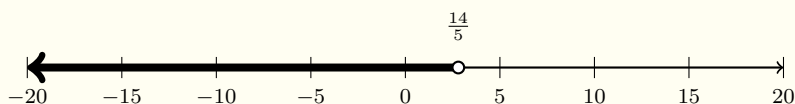
$$\frac{3x+2}{6} + \frac{2(x-5)}{6} < 1 \Rightarrow \frac{3x+2+2x-10}{6} < 1 \Rightarrow \frac{5x-8}{6} < 1.$$

Multiply by 6 (positive):

$$5x - 8 < 6 \Rightarrow 5x < 14 \Rightarrow x < \frac{14}{5} = 2.8.$$

Step 2 (interval form). $\left(-\infty, \frac{14}{5}\right)$.

Step 3 (number line).



Worked Example 17

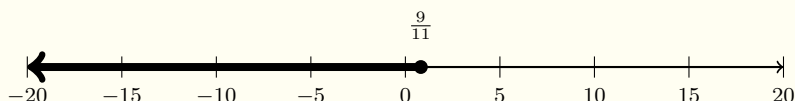
Solve $-2(1 - 3x) \leq 7 - 5x$.

Step 1 (algebra). Expand: $-2(1 - 3x) = -2 + 6x$. So

$$-2 + 6x \leq 7 - 5x \Rightarrow 11x \leq 9 \Rightarrow x \leq \frac{9}{11} \approx 0.818.$$

Step 2 (interval form). $\left(-\infty, \frac{9}{11}\right]$.

Step 3 (number line).



Worked Example 18

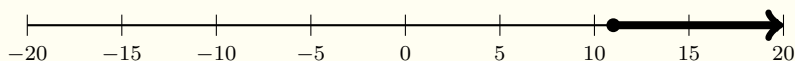
Solve $0.4x - 1.6 \geq 2.8$.

Step 1 (algebra).

$$0.4x \geq 2.8 + 1.6 = 4.4 \Rightarrow x \geq \frac{4.4}{0.4} = 11.$$

Step 2 (interval form). $[11, \infty)$.

Step 3 (number line).



3 Compound Inequalities

Worked Example 19

Solve $-3 \leq 2x - 1 < 7$.

Step 1 (algebra). Add 1 to all three parts:

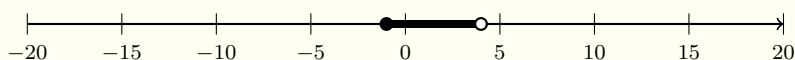
$$-2 \leq 2x < 8.$$

Divide by 2:

$$-1 \leq x < 4.$$

Step 2 (interval form). $[-1, 4)$.

Step 3 (number line).



Worked Example 20

Solve $4 < 3x + 2 \leq 14$.

Step 1 (algebra). Subtract 2:

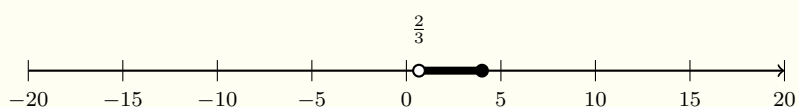
$$2 < 3x \leq 12.$$

Divide by 3:

$$\frac{2}{3} < x \leq 4.$$

Step 2 (interval form). $\left(\frac{2}{3}, 4\right]$.

Step 3 (number line).



Worked Example 21

Solve $-10 < 5 - 3x \leq 2$.

Step 1 (algebra). Subtract 5:

$$-15 < -3x \leq -3.$$

Divide by -3 (negative, flip both inequalities):

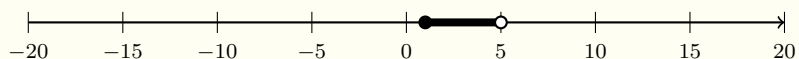
$$5 > x \geq 1.$$

Rewrite in increasing order:

$$1 \leq x < 5.$$

Step 2 (interval form). $[1, 5)$.

Step 3 (number line).



Worked Example 22

Solve $\frac{1}{2} \leq \frac{x+3}{4} < 2$.

Step 1 (algebra). Multiply by 4 (positive):

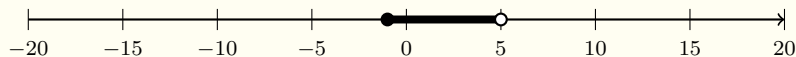
$$2 \leq x + 3 < 8.$$

Subtract 3:

$$-1 \leq x < 5.$$

Step 2 (interval form). $[-1, 5)$.

Step 3 (number line).



Worked Example 23

Solve $-2 \leq 3(1 - x) < 10$.

Step 1 (algebra). Expand: $3(1 - x) = 3 - 3x$.

$$-2 \leq 3 - 3x < 10.$$

Subtract 3:

$$-5 \leq -3x < 7.$$

Divide by -3 (negative, flip both):

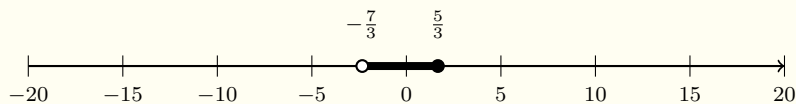
$$\frac{5}{3} \geq x > -\frac{7}{3}.$$

Rewrite in increasing order:

$$-\frac{7}{3} < x \leq \frac{5}{3}.$$

Step 2 (interval form). $\left(-\frac{7}{3}, \frac{5}{3}\right]$.

Step 3 (number line).



Worked Example 24

Solve $1 \leq \frac{2x-1}{3} \leq 5$.

Step 1 (algebra). Multiply by 3:

$$3 \leq 2x - 1 \leq 15.$$

Add 1:

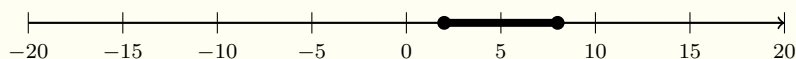
$$4 \leq 2x \leq 16.$$

Divide by 2:

$$2 \leq x \leq 8.$$

Step 2 (interval form). $[2, 8]$.

Step 3 (number line).



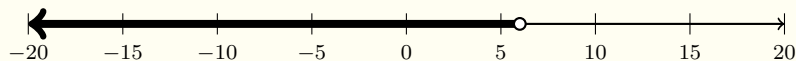
4 Convert Between Inequality form and Interval Notation

Worked Example 25

Given interval $(-\infty, 6)$.

Inequality form. $x < 6$.

Number line.

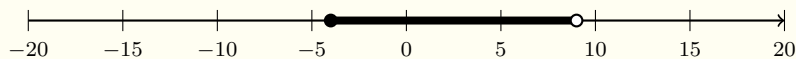


Worked Example 26

Given interval $[-4, 9)$.

Inequality form. $-4 \leq x < 9$.

Number line.

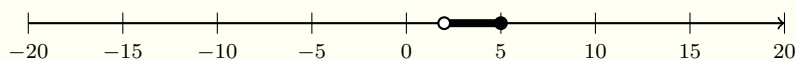


Worked Example 27

Given interval $(2, 5]$.

Inequality form. $2 < x \leq 5$.

Number line.

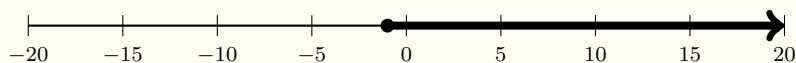


Worked Example 28

Given interval $[-1, \infty)$.

Inequality form. $x \geq -1$.

Number line.

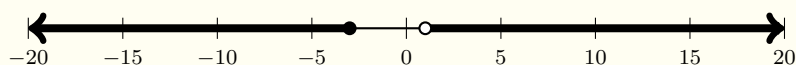


Worked Example 29

Given interval $(-\infty, -3] \cup (1, \infty)$.

Inequality form. $x \leq -3$ or $x > 1$.

Number line.

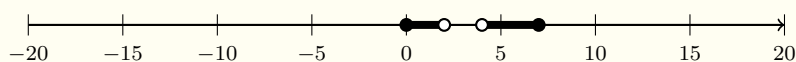


Worked Example 30

Given interval $[0, 2) \cup (4, 7]$.

Inequality form. $0 \leq x < 2$ or $4 < x \leq 7$.

Number line.



End of solution.